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Indoor Air Quality at the Climate-Health Nexus: The Case for Enforceable Standards

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The COVID-19 pandemic exposed a longstanding governance gap: most people spend much of their time indoors, yet indoor air quality remains far less regulated than outdoor air. Climate change is now worsening that gap by increasing the frequency and severity of wildfire smoke and heat-driven building closures, both of which degrade air quality in homes, schools, workplaces and other occupied spaces. Poor indoor air quality is therefore no longer only a building-performance issue; it is a climate-health problem with clear implications for infection control, chronic disease, cognitive performance and equity. The case for action is strengthened by growing evidence that the costs of inaction are substantial, while the benefits of better indoor environments are measurable. This perspective piece argues that indoor air quality should be governed through enforceable, performance-based standards with statutory force, beginning in settings where exposure is unavoidable and vulnerability is high. Australia is used as an illustrative case to show how monitoring, institutional responsibility and phased implementation could be built into a practical regulatory pathway.

A neglected environment

Indoor air quality remains a neglected area of public health and environmental governance (Kennedy et al., 2025). By comparison with outdoor air, which is supported by relatively mature regulatory systems, indoor environments are often treated as passive rather than active settings that require management. That assumption is no longer tenable. The COVID-19 pandemic made clear that poorly ventilated indoor spaces can amplify airborne transmission of infectious disease (Morawska and Milton, 2020), and the years since have brought ventilation and indoor air quality into public policy discussions in parts of Europe, North America and Asia (Morawska, Li and Salthammer, 2024). The challenge now is to sustain that attention beyond pandemic preparedness and apply it to the broader indoor exposures intensified by climate change.

Climate change and indoor air quality

Climate change affects indoor air quality through several pathways. Wildfire smoke infiltrates buildings and raises indoor particulate concentrations even where occupants assume they are protected. When outdoor air is hot or polluted, people often seal buildings to keep heat or smoke out, which can trap pollutants generated indoors, including carbon dioxide, volatile organic compounds and biological contaminants. The drive toward more airtight and energy-efficient buildings can amplify pollutant accumulation if ventilation and filtration requirements are not simultaneously updated. In short, indoor environments cannot be assumed to be a safe refuge from outdoor air pollution.

The wildfire pathway is the most visible and most urgent, with recent extreme burns in Australia during 2019 and, most recently, across Europe and North America in 2026. Climate change is projected to increase smoke exposure in many regions, including places that historically experienced fewer large fires (Ford et al., 2018); more than two billion people are already exposed to at least one day of substantial fire-sourced air pollution each year, with the burden falling disproportionately on low- and middle-income countries (Xu et al., 2023). Household combustion of solid fuels for cooking and heating is already a dominant source of indoor PM_{2.5} and accounts for an estimated 3.2 million deaths annually (WHO, 2022), and in low- and middle-income countries this burden remains a major source of premature mortality even before wildfire smoke adds a further layer of risk on top of it. Even in high-income settings, studies show that buildings provide only partial protection during smoke events, especially in older housing and in lower-income communities that have fewer options for filtration, relocation or reduced exposure (Liang et al., 2021). The public health implication is straightforward: telling people to shelter indoors is not enough unless indoor air itself is protected.

A second pathway is behavioral sealing. During heat waves or smoke events, occupants close windows and doors to reduce outdoor infiltration. That response can be partially helpful, but without mechanical ventilation or filtration it also allows indoor pollutants to accumulate, and the protection it offers against fine particulate matter is often only partial (Reisen et al., 2019). Any indoor air quality standards must ensure that in reducing one hazard, others are not inadvertently made worse.

The final pathway is the energy-efficiency trade-off. As buildings become more airtight to reduce energy demand and improve climate performance (CSIRO, 2024), the health consequences of insufficient ventilation become harder to ignore (Sundell et al., 2011). This does not mean energy efficiency is the problem; rather, it means energy and indoor air policy must be designed together. Buildings that reduce emissions should not do so by worsening the quality of the air inside them. None of these pathways are one-off events, and standards built for average conditions will increasingly fail exactly when they are needed most.

The economic imperative

Poor indoor air quality imposes costs through healthcare expenditure, lost productivity and broader macroeconomic losses (World Bank and IHME, 2016; Fisk and Rosenfeld, 1997). The evidence is strongest for respiratory and cardiovascular outcomes (Brook et al., 2010; Du, Y. et al., 2020; Maung et al., 2022), but it also extends to cognition (MacNaughton et al., 2017; Allen et al., 2016), absenteeism and reduced work performance (Palacios, Eichholtz and Kok, 2020). While individual estimates vary, the direction of effect is consistent: better indoor environments are associated with better health and higher productivity. An early Australian estimate put the cost of poor indoor air quality to the national economy at AUD 12 billion annually (Brown, 1998); given the scale of change since then, including the COVID-19 pandemic and the intensification of bushfire activity, that figure is best read as an order-of-magnitude indication rather than a current national estimate, since no methodologically comparable update yet exists - itself a telling sign of how little policy priority the issue has received.

Healthcare costs are driven by disease linked to long-term exposure to fine particulate matter and other pollutants found both indoors and outdoors; the clearest quantified mortality and disease burden estimates come from ambient PM_{2.5} exposure (Pope et al., 2002; Murray et

al., 2020), and the same biological pathways are implicated for the indoor PM2.5 that people are exposed to for most of the day. Indoor PM2.5 matters because ambient standards do not protect against indoor sources such as cooking, gas appliances and infiltration from outside. Nitrogen dioxide (NO2) is also an important indoor pollutant, particularly in homes with gas cooking or unflued gas heating. These combustion sources can produce short-term indoor NO2 concentrations that are relevant to respiratory health; epidemiological evidence links gas cooking and indoor NO2 exposure with asthma and wheeze in children, although estimates of causality and attributable burden remain contested (Lin, et al., 2013; Kashtan et al., 2024). Household air pollution remains a major global burden, and the burden is concentrated in environments where regulation has historically been weakest.

Productivity losses arise through both immediate cognitive effects and increased illness-related absence (Allen et al., 2016). Controlled studies have shown better cognitive performance in low-VOC and well-ventilated environments than in conventional indoor settings. Elevated CO2 concentrations, which are common in classrooms, offices and meeting rooms, are associated with poorer decision-making and slower cognitive performance (Du, B. et al., 2020). These findings matter because even small changes in cognitive function can have large cumulative effects across workplaces and schools. The costs of inaction are likely to significantly exceed the costs of monitoring and regulation, even if the precise magnitudes differ across countries and building types.

The emerging regulatory landscape

Indoor air quality standards are still limited in most national regulatory systems (Dimitroulopoulou et al., 2023). Where guidance exists, it is often advisory rather than enforceable, and thresholds vary widely across jurisdictions. The World Health Organization has strengthened recommendations on particulate matter and highlighted ventilation as a public health issue, but its guidance is not binding (WHO, 2021a). A parallel WHO roadmap developed during the pandemic called for ventilation standards in public buildings, but it too lacked enforcement mechanisms at the national level (WHO, 2021b). At the same time, the policy landscape is beginning to shift. The European Union has tightened outdoor air quality rules (European Commission, 2024), the US' CDC has issued indoor air guidance (CDC, 2023), and a small number of countries have begun integrating indoor air monitoring into schools and public buildings (Morawska et al., 2024). Health entered climate negotiations more directly at COP30 through the Belem Health Action Plan, which committed states to strengthen health-system resilience to air quality related risks (WHO, 2025), while a parallel Global Pledge for Healthy Indoor Air had drawn more than 200 signatories within its first year (Global Commission on Healthy Indoor Air, 2025).

These developments do not yet amount to a coherent international regime, but they do show a growing recognition that indoor air quality is a distinct policy domain. They also strengthen the case for a more explicit legal and regulatory framework. The most promising approach is performance-based regulation: standards that measure actual conditions in occupied spaces rather than relying only on fixed design assumptions that a warming climate makes less safe every year. Such standards are more likely to protect health across different building types, occupancy patterns and climates, because they measure what is actually happening rather than what was assumed at the design stage.

Illustrative national progress: Australia

Australia is a useful example because it combines growing wildfire smoke exposure, a federal system that fragments responsibility across jurisdictions, and an emerging research and policy community focused on indoor air quality. The main challenge is that the evidence base remains thin. Recent estimates suggest that less than 0.03% of Australian buildings have been systematically monitored for indoor air pollution, making regulation difficult without first improving data collection (Miller and Morawska, 2025). That said, progress is visible. State-level initiatives in schools and public buildings show that low-cost monitoring and ventilation improvements can deliver measurable gains (Cooper et al., 2025; Majumdar et al., 2026). These programmes suggest a plausible sequence for reform: build the evidence base, develop institutional capacity, and then introduce performance-based standards in settings where exposure is routine and vulnerable groups are present. A staged approach is especially sensible in schools and healthcare facilities, where the duty of care is clear and the potential benefits are highest - a sequencing also recommended in expert submissions to the New South Wales Legislative Council's inquiry into clean indoor air (Johnston, 2026).

Australia is not unique in this respect, though the lessons of a federal system with fragmented responsibility will not transfer directly to unitary states, or to the low- and middle-income countries where household fuel combustion, not building codes, is the dominant exposure pathway. Elsewhere, some countries have introduced school-based CO₂ monitoring, indoor air action plans, or statutory requirements for public spaces (UK Department for Education, 2026; Belgian Government, 2022). Finland has run a government-mandated Indoor Air and Health Programme since 2018 (Finnish Institute for Health and Welfare, 2018), while China's GB 50736 standard sets minimum air supply rates for civil buildings, though compliance verification remains limited, illustrating the gap between standard-setting and enforcement that performance-based regimes are designed to close (Standardization Administration of China, 2012). These examples show that the policy direction is moving from voluntary guidance toward more defined obligations, even if full enforcement remains uncommon.

Implementation gaps and requirements for effective standards

Translating momentum into enforceable standards requires addressing several barriers: governance fragmentation, monitoring and design challenges, and equity risks. None of these barriers is insurmountable, but all need to be addressed explicitly.

The responsibility for buildings, health, environment and energy is usually spread across multiple agencies. That fragmentation makes it easy for indoor air quality to be no one agency's specific responsibility. A workable regime needs a clearly designated authority, or at minimum a formal mechanism for coordination, so that standards are not merely symbolic.

The standards must be measurable and affordable. Compliance costs are uneven: it is more expensive to retrofit older buildings than to include ventilation and filtration in new construction, and in rented housing the landlord who pays rarely benefits directly. That argues for phased implementation, targeted support for vulnerable building stock and a monitoring infrastructure that precedes or accompanies regulation. Outcome-based standards are only credible if building operators have practical ways to measure and improve performance.

And standards should be designed to reduce inequity rather than widen it. Vulnerable populations often have the least control over their indoor environments and the fewest resources to adapt. That makes schools, healthcare facilities and other public buildings

sensible first targets, both because people cannot easily avoid exposure there and because compliance can be publicly accountable. In federal systems, national harmonisation may be difficult, but it is still preferable to a patchwork of weak or inconsistent rules.

Towards a right to clean indoor air

The case for enforceable indoor air quality standards has scientific, economic and normative foundations. Scientifically, indoor pollution affects respiratory health, cardiovascular risk, cognition and productivity at concentrations that are common in ordinary buildings. Economically, the costs of inaction in healthcare and lost productivity are likely to exceed the costs of monitoring and regulation, even though more formal cost-benefit evidence is still needed. The third foundation is more aspirational than the first two, but no less relevant to how standards should ultimately be framed: in 2022, the United Nations General Assembly formally recognised access to a clean, healthy and sustainable environment as a human right (United Nations General Assembly, 2022), and the UN Special Rapporteur on the human right to a clean, healthy and sustainable environment has since identified air pollution, indoor as well as outdoor, as a direct threat to that right (Puentes Riano, 2026). That recognition strengthens the claim that governments have a duty to prevent avoidable indoor exposure, and gives advocates a normative benchmark to hold governments to even before statutory standards exist.

What makes the current moment important is not a sudden discovery of the problem, but the convergence of evidence, public awareness and policy opportunity. Post-pandemic concern about ventilation, climate-amplified smoke exposure and the gradual emergence of monitoring and standards in some jurisdictions have created conditions that did not exist a decade ago. The policy implication is not that every building should be regulated in the same way immediately, but that governments should establish statutory authority, begin with high-priority public settings, mandate monitoring, and align indoor air standards with energy policy so that climate mitigation and health protection advance together. Electrifying cooking and, alongside efficient ventilation and filtration, can provide co-benefits by reducing indoor combustion pollutants and associated inequities while also reducing fossil fuel use and greenhouse gas emissions. These benefits will be greatest where policy support enables low-income households and older buildings to participate rather than leaving them with the highest exposures and fewest options for improvement.

The question is no longer whether indoor air quality matters. It does. The question is whether institutions will create enforceable standards soon enough to prevent avoidable harm.

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